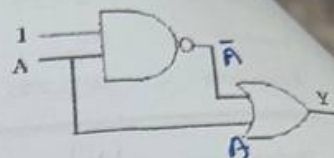




Problem # 1: Answer the following Multiple-choice questions: [12 x 1pt]

- 1- The output Y of the combinational circuit shown to the right is:



- A. 0 ☒ B. 1 C. A D. $\bar{A}$
- 2- The range of signed decimal numbers that can be represented by 7 bits is:

- A. -31 to +31 B. -63 to +63 ☒ C. -64 to +63 D. -127 to +127

- 3- The octal representation of $(156)_{10}$ is:

- A. $(231)_8$ B. $(234)_8$ C. $(236)_8$ D. $(232.01)_8$

- 4- In 16-bit 2's complement representation, the decimal number -28 is:

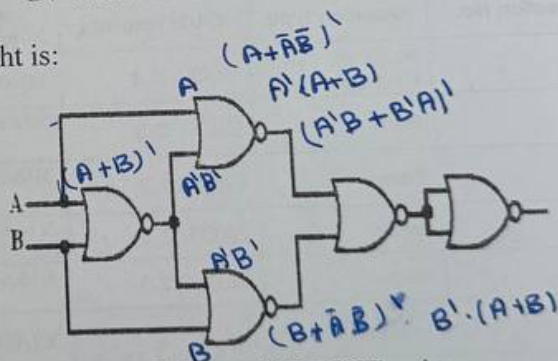
- A. 00000000000011100 B. 11111111111100100 ☒ C. 11111111111101100 D. 00000000000011100

- 5- The result of $(F0A00002)_{16} - (89D34FA9)_{16}$ is:

- A. $(666CB0F9)_{16}$ B. $(70ACB0F9)_{16}$ C. $(766CB0F9)_{16}$ D. None of the above

- 6- The combinational circuit shown in the right is:

- A. $\bar{A}\bar{B} + AB$ B. $\bar{A} + \bar{B}$ ☒ C. $\bar{A}B + A\bar{B}$ D. $\bar{A} \cdot \bar{B}$



- 7- The result of adding the hexadecimal numbers $(2C5AFF)_{16} + (1D7C67)_{16}$ is:

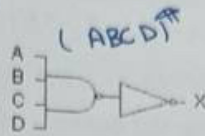
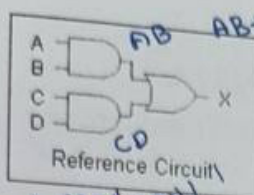
- A. $(49D226)_{16}$ B. $(43E567)_{16}$ ☒ D. None of the above

- 8- Using 6-bit signed numbers (two's complement), which of the following decimal additions causes overflow?

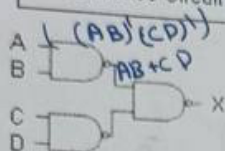
- A. $15 + 10$ B. $-20 + 5$ ☒ D. $-15 + -20$



9. Which of the figures (A to D) is equivalent to the reference circuit



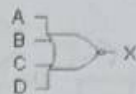
(A)



(B)



(C)



(D)

A. (A)

C. (C)

B. (B)

D. (D)

7 0 1
111 000 001

10. The equivalent binary representation of the following octal number $(701)_8$ is:

A. $(111000101)_2$

B. $(101001001)_2$

C. $(111000001)_2$

D. $(011001001)_2$

11. If $F(x, y, z, w) = \pi(1, 2, 4, 8, 13, 14, 15)$, the complement of F ($\bar{F}$) is:

A. $\pi(0, 3, 5, 6, 7, 9, 10, 11, 12)$

B. $\sum(1, 2, 4, 8, 13, 14, 15)$

C. $\sum(0, 3, 5, 6, 7, 9, 10, 11, 12)$

D. $\pi(1, 2, 4, 8, 13, 14, 15)$

12. The equivalent hexadecimal value of the binary number $(1001110101010)_2$ is:

A. $(13AA)_{16}$

B. $(9D50)_{16}$

C. $(47AB)_{16}$

D. $(9CA0)_{16}$

*** Write the correct answers to problem # 1 in the table below:

Q#1	Q#2	Q#3	Q#4	Q#5	Q#6	Q#7	Q#8	Q#9	Q#10	Q#11	Q#12
B	C	B	B	A	C	D	D	B	C	C	A

A

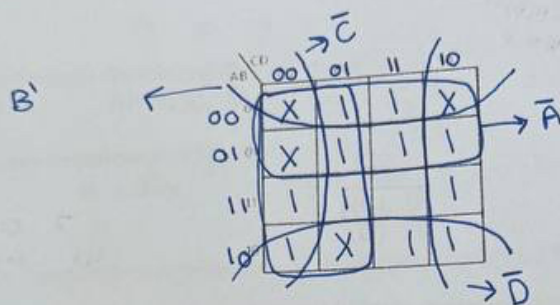


Problem #2: Find the most simplified SOP expression for

$$F(A, B, C, D) = \sum(1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14)$$

together with the don't care conditions $d(A, B, C, D) = \sum(0, 2, 4, 9)$.

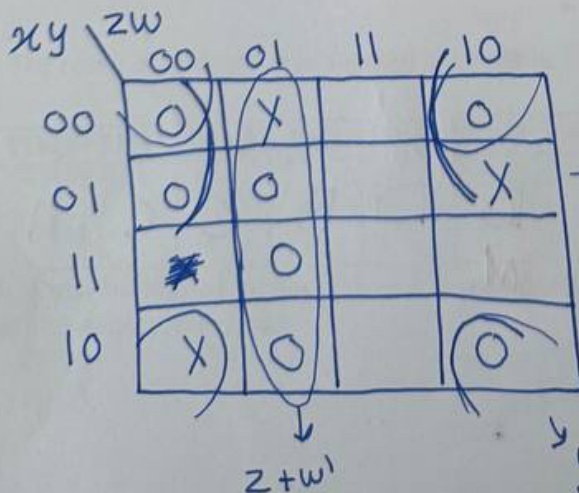
You must show all steps including the drawing of the cells and grouping cells together, etc. [5pts]



$$F(A, B, C, D) = \bar{A} + \bar{B} + \bar{C} + \bar{D}$$

Problem #3: The function $F(x, y, z, w) = \pi(0, 2, 4, 5, 9, 10, 13)$ is simplified to

$F(x, y, z, w) = (x + w) \cdot (z + \bar{w}) \cdot (y + w)$. Are there any don't care conditions? If yes, what are they? [3pts]



$$\begin{aligned} (x'w')' &\rightarrow x+w \\ (z'w)' &\rightarrow z+w' \\ (y'w)' &\rightarrow y+w \end{aligned}$$

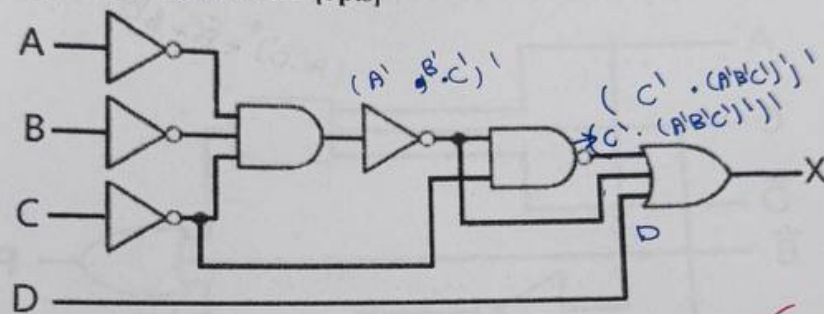
$$\rightarrow x+w$$

yes

$$d(x, y, z, w) = \sum m(1, 6, 8)$$



Problem # 4: Based on the below combinational circuit, find the minimal form of X function using Boolean algebra rules. [5pts]



$$\begin{aligned}
 X &= (D + (A'B'C')' + (C' \cdot (A'B'C')'))' \\
 X &= D + (A+B+C) + (C' \cdot (A+B+C))' \\
 X &= D + A + B + C + (C + A'B'C') \\
 X &= A + B + C + D + C + A'B'C' \\
 X &= A + B + C + D + A'B'C'
 \end{aligned}$$

+ 2.5

+ 1.5



Problem # 5: Implement the below combinational circuit using only NAND gates. [5pts]

